



Dual-hierarchical knowledge distillation for video captioning

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ABSTRACT

Video captioning remains a challenging task due to the tradeoff between accuracy and computational efficiency. We propose **CapDistill**, a dual hierarchical distillation framework that transfers semantic knowledge from a powerful teacher model to a lightweight student model. CapDistill captures object-level and action-level semantics from captions and transfers multilevel knowledge including object features, action features, and word-level predictions through a hierarchical strategy. To reduce the impact of noisy annotations, we introduce a caption quality grading mechanism that assigns quality-based weights to training captions. Experiments on MSR-VTT and MSVD demonstrate that CapDistill achieves state-of-the-art accuracy while significantly reducing inference cost. Code is available at: <https://github.com/ccc000-png/SFTCap>.

1. Introduction

Video captioning, which involves generating coherent natural language descriptions from video content, plays a crucial role in applications such as assistive technologies for the visually impaired [1], content-based video retrieval [2], and autonomous systems [3]. Unlike image captioning that deals with static content, video captioning must capture temporal dynamics, object interactions, and scene transitions. This requires advanced visual representations, effective temporal modeling, and precise alignment between visual and linguistic modalities.

Early methods evolved from template-based pipelines to encoder-decoder architectures using Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), such as Long Short-Term Memory networks (LSTMs) and Gated Recurrent Units (GRUs) [4]. The adoption of attention mechanisms allowed models to dynamically attend to informative temporal segments, improving alignment between visual input and generated text [5]. More recently, transformer-based models have replaced RNNs, leveraging multi-head self-attention for superior temporal and contextual modeling [6]. Progress has also been driven by multimodal fusion techniques (e.g., subtitles, audio, and Automatic Speech Recognition (ASR)) [7], graph-based reasoning for modeling object-action relationships [8,9], and reinforcement learning for metric-aware optimization [10].

Despite this progress, three major challenges persist:

- **Efficient temporal modeling:** Modeling long-range dependencies across frames is crucial for understanding complex video content but

is often computationally expensive, limiting deployment in real-time or resource-constrained environments.

- **Annotation variability and quality:** Although datasets like MSVD [11] and MSR-VTT [12] provide diverse human-written captions, not all annotations are equally informative. Some captions include vague descriptions, subjective phrasing, or misaligned semantics that weaken the supervision signal during training. For example, as shown in Fig. 1, “a man is standing with road” introduces off-topic content, while “a girl is jumping” may misrepresent a water-skiing scene. While these captions are linguistically valid, they hinder consistency in model learning and semantic alignment.
- **Semantic feature learning:** Generating high-quality captions requires structured understanding of objects, actions, and spatiotemporal relationships. Efficiently learning such features in a compact representation remains an open problem.

To address these challenges, we propose a dual-hierarchical distillation framework that incorporates a caption evaluation mechanism to prioritize semantically informative annotations without discarding valid linguistic diversity. We opted for knowledge distillation from the out-set because it offers an effective strategy to leverage the strong representational capacity of large teacher models while enabling efficient compression into smaller student models suitable for real-world deployment. In video captioning, this approach allows the model to capture fine-grained semantic and temporal dependencies without requiring prohibitively large architectures. It also facilitates more stable and sample-efficient training compared to training compact models from

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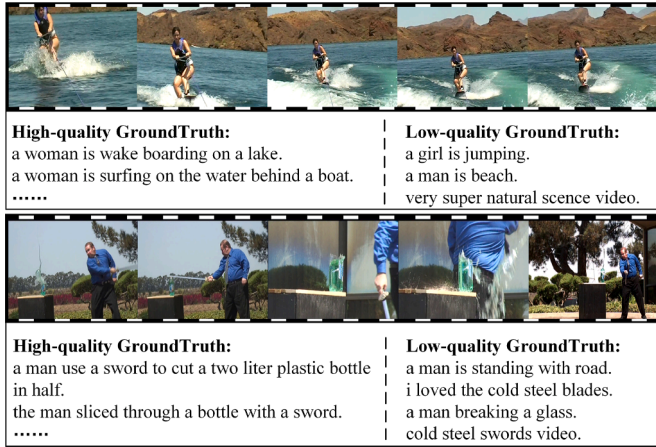


Fig. 1. Examples of annotation variability in the MSVD dataset. Some captions, such as “a girl is jumping,” misrepresent a water-skiing scene, while others, like “a man is standing with road,” introduce off-topic content. Although linguistically valid, such captions hinder semantic alignment and consistent model learning.

scratch, which is especially important when annotations are noisy or limited.

Unlike existing methods that rely on end-to-end training with uniform annotation treatment, our approach introduces a two-stage distillation process. First, high-quality captions are used to guide hierarchical learning in the teacher model by distilling object-level and action-level semantics. Second, the learned multi-level knowledge (including object features, action features, and word-level predictions) is transferred from the teacher to a lightweight student model. To the best of our knowledge, CapDistill is the first framework to realize this dual-hierarchical distillation paradigm for video captioning. The integration of caption grading further enhances training robustness by assigning adaptive weights to noisy or inconsistent annotations.

The main contributions of this work are:

- **Efficient temporal modeling through distillation:** We adopt a teacher-student framework in which a high-capacity teacher captures rich temporal and contextual video features, which are distilled into a compact student model to enable efficient and scalable inference.
- **Noise-robust training via caption grading:** A caption quality grading mechanism is introduced to assess and weight ground-truth captions based on their semantic relevance and consistency, reducing the influence of noisy or ambiguous annotations during supervision.
- **Dual-hierarchical distillation of semantic knowledge:** Our framework jointly distills object-level and action-level semantics, first from captions to the teacher network, then from the teacher to the student model. This hierarchical supervision enhances semantic alignment between visual inputs and linguistic outputs.

In summary, CapDistill addresses key challenges in video captioning through efficient modeling, noise-resilient supervision, and dual-hierarchical knowledge transfer. Extensive experiments on MSVD, MSR-VTT and VATEX [13] demonstrate that our method outperforms state-of-the-art approaches while maintaining scalability and generalization to real-world scenarios.

2. Related works

2.1. Video captioning

Early video captioning methods used template-based approaches that mapped detected subjects, verbs, and objects (SVO) into predefined

sentence structures. These were later replaced by deep learning-based models, which demonstrated improved capabilities to capture complex spatio-temporal semantics and generate context-aware descriptions.

Encoder-decoder architectures. Encoder-decoder frameworks have played a central role in advancing video captioning. In these models, Convolutional Neural Networks (CNNs) were commonly employed for spatial feature extraction, while Recurrent Neural Networks (RNNs) were initially used to model temporal dependencies. Later works introduced attention mechanisms and graph-based reasoning modules to improve temporal alignment and capture inter-object relations. For instance, Res-F2F [14] fused multi-stream CNN features, and VASTA [5] applied adaptive spatio-temporal attention for informative frame selection. HTG + HMG [8] and KG-VCN [9] utilized graph structures to model complex visual semantics, while IVRC [15] proposed a token shift module and Refineformer for fine-grained temporal modeling. MAN [16] introduced a memory-augmented encoder and decoder with separate visual and textual memories to capture implicit external knowledge within the dataset. GSEN [17] and Track4Cap [18] emphasized global semantics and object interactions to enhance caption coherence. Collectively, these methods improved contextual understanding and semantic fidelity in video captioning.

Syntax-aware models. Syntax-aware methods aimed to enhance linguistic quality by incorporating grammatical and semantic structure into caption generation. SAAT [19] focused on verb-centric alignment to better capture actions, while TTA [20] aligned object tags with descriptive tokens to improve visual-language grounding. RSFD [21] employed frequency-aware attention to emphasize rare but informative words. SNCL [22] introduced hierarchical reasoning guided by semantic contrastive learning. MesNet [4] adopted a multi-layer GRU decoder, where the lower layers modeled syntax and the higher layers encoded semantics. EmVidCap [23] further extended this line by incorporating an emotion-aware framework and dataset, combining factual and emotional streams to generate more expressive and engaging captions.

Retrieval and copy mechanisms. Retrieval-based methods enhanced caption diversity and fluency by incorporating external corpora. OpenBook [2] retrieved contextually similar captions and integrated selected phrases using a copy mechanism. CARE [24] improved concept detection through multimodal video-to-text retrieval and applied global-local semantic guidance to generate informative and coherent descriptions. These methods benefited from external knowledge, but were highly dependent on the relevance and quality of the retrieved content.

Non-autoregressive and action-aware models. ALSO-Net [25] enhanced non-autoregressive video captioning by integrating high-level action features and optimizing motion-related losses, improving verb accuracy and coherence with lower complexity.

End-to-end transformer models. Transformer-based models unified spatial, temporal, and semantic modeling through self-attention. Models such as SwinBERT [6] and CoCap [26] replaced RNNs with transformer blocks, enabling joint optimization of vision and language features. Although these approaches achieved strong performance, they incurred significant computational costs and reduced flexibility for modular adaptation.

2.2. Training techniques

Advanced training paradigms have been explored to improve generalization, efficiency, and metric alignment.

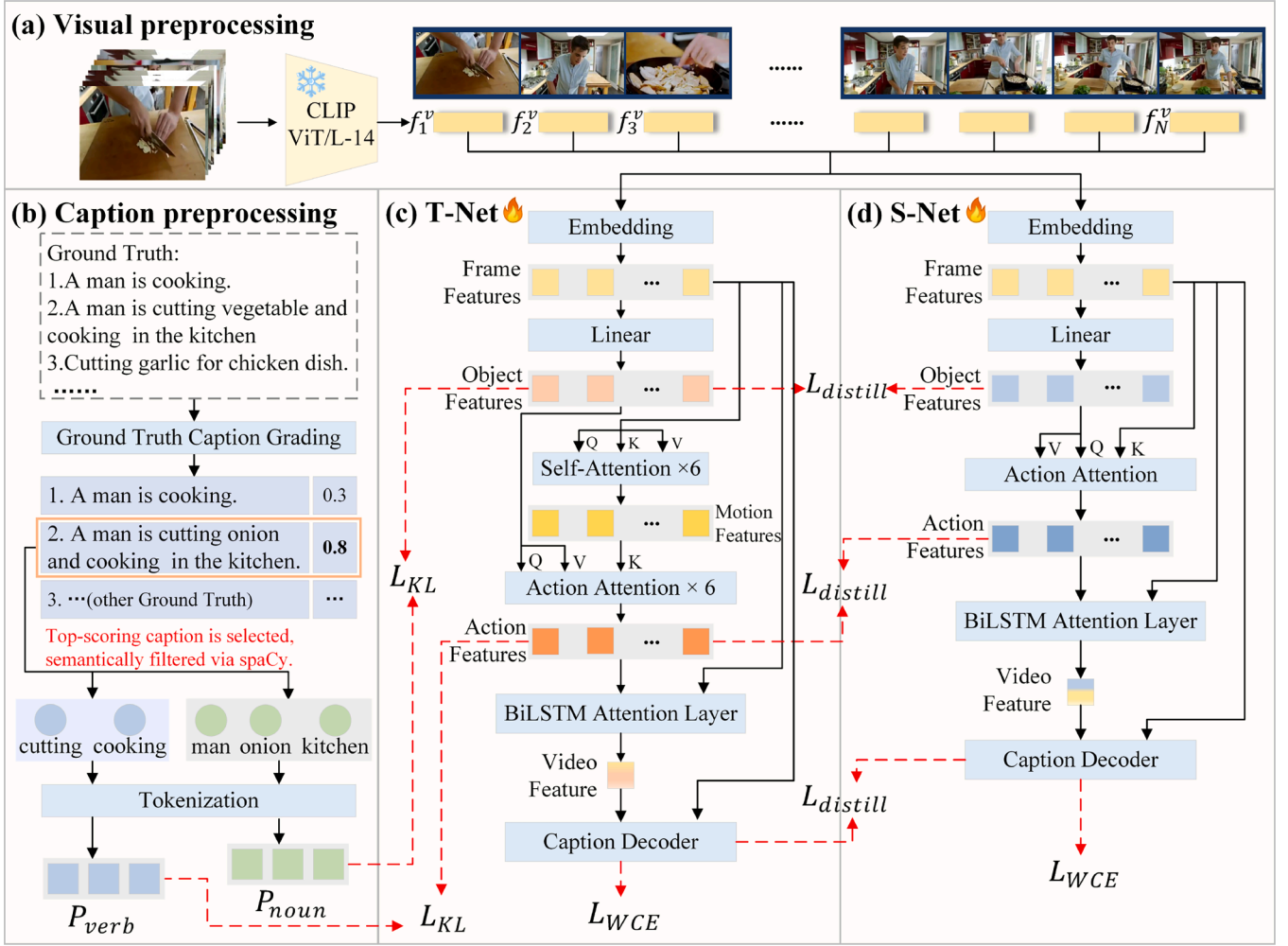


Fig. 2. Overview of the proposed CapDistill framework with dual hierarchical knowledge distillation. During training, ground-truth captions are first graded. The resulting grade is used as a soft weight in the weighted cross-entropy loss L_{WCE} , enabling the model to prioritize more informative captions while preserving linguistic diversity. Additionally, the highest-scoring caption per video is used to supervise hierarchical feature learning in the teacher network, guiding object- and action-level semantic representation. During inference, only the lightweight student network is used to generate captions, achieving efficient inference with strong performance.

Reinforcement learning. Methods like DRPN [10] optimized sequence-level captioning metrics (e.g., CIDEr [27], BLEU [28]) using reward signals. Although effective, such methods often introduced instability and required carefully crafted reward functions.

Adversarial training. Adversarial learning frameworks for video captioning [29] employed a generator-discriminator setup, where the discriminator guided the generator to produce more natural and fluent captions. Although this approach improved linguistic quality, it introduced additional training complexity and required careful balancing between adversarial and language modeling objectives.

Contrastive learning. Contrastive methods such as EMCL [30] aligned visual and textual representations in shared latent spaces by maximizing similarity for paired inputs. While efficient, they sometimes sacrificed fine-grained detail due to representation compression.

Knowledge distillation. Knowledge distillation aims to transfer knowledge from a high-capacity teacher model to a compact student model. Methods such as STG-KD [31] and ORG-TRL [32] distilled temporal or relational representations but were typically limited to single-level features, which constrained the semantic diversity transferred to the student.

3. Methodology

3.1. Overview of the proposed method

The proposed video captioning framework, **CapDistill**, addresses the trade-off between computational efficiency and semantic richness via a teacher-student architecture with *dual hierarchical knowledge distillation*. As shown in Fig. 2, the framework comprises four main components: visual preprocessing, caption preprocessing, a teacher network (T-Net), and a student network (S-Net).

The visual preprocessing module extracts frame-level features using a pre-trained CLIP-ViT model [33], which are shared by both T-Net and S-Net to ensure consistent visual input. T-Net encodes multi-level semantic representations, while S-Net learns to approximate these through hierarchical distillation.

The caption preprocessing module assesses ground-truth caption quality using a CIDEr-based metric. These scores are used in two ways: (1) to weight the cross-entropy loss for both T-Net and S-Net, and (2) to guide T-Net's object and action feature learning using only the highest-quality captions. This reduces the influence of noisy or inconsistent annotations.

At inference time, only the lightweight S-Net is used to generate captions, significantly lowering computational cost while preserving

semantic fidelity. This makes CapDistill well suited for real-time and resource-constrained video captioning scenarios.

3.2. Visual preprocessing: Frame-level feature extraction

Unlike prior methods that rely on computationally expensive 3D CNNs such as I3D [34] or C3D [35], our approach uses 2D frame-level features for efficiency. Given a video, we randomly sample N frames $\{I_1, I_2, \dots, I_N\}$. Each frame I_n is passed through the CLIP visual encoder to extract a feature vector:

$$f_n^v = \text{CLIP}(I_n). \quad (1)$$

The resulting set of features is $F_v = \{f_1^v, \dots, f_N^v\}$, where $f_n^v \in \mathbb{R}^d$ and $d = 512$. These features are used by both the teacher network (T-Net) and the student network (S-Net).

3.3. Caption preprocessing: Ground-truth caption grading

Video captioning datasets like MSVD and MSR-VTT offer diverse human-written captions per video, but vary in quality. We introduce a caption grading mechanism that scores each caption based on (1) semantic relevance via average TF-IDF and (2) reference consistency via CIDEr similarity with other captions of the same video. The summed score is normalized per video and used as a soft weight during training to prioritize informative captions. The highest-scoring caption also supervises hierarchical feature learning in the teacher network.

Semantic relevance via TF-IDF: We compute the Term Frequency-Inverse Document Frequency (TF-IDF) for each word w in a caption c as:

$$\text{TF-IDF}(w, c) = \text{TF}(w, c) \times \text{IDF}(w), \quad (2)$$

where

$$\text{IDF}(w) = \log \left(\frac{M}{1 + \text{DF}(w)} \right), \quad (3)$$

M is the number of captions in the MSCOCO validation set, and $\text{DF}(w)$ is the number of captions containing word w .

Reference consistency via CIDEr: To evaluate consensus, we treat each ground-truth caption c as a candidate and use the remaining captions for the same video as reference sentences. The CIDEr score is computed as:

$$\text{CIDEr}(c) = \frac{1}{4} \sum_{n=1}^4 \frac{1}{m} \sum_{j=1}^m \frac{\sum_{g \in G} \text{TF-IDF}(g, c) \cdot \text{TF-IDF}(g, r_j)}{\sqrt{\sum_{g \in G} \text{TF-IDF}(g, c)^2} \cdot \sqrt{\sum_{g \in G} \text{TF-IDF}(g, r_j)^2}}, \quad (4)$$

where G is the set of n -grams and m is the number of reference captions $\{r_j\}$.

Caption quality score and selection: The overall quality score is defined as:

$$S(c) = \frac{1}{|W|} \sum_{w \in W} \text{TF-IDF}(w, c) + \text{CIDEr}(c), \quad (5)$$

where W is the set of words in c . Scores are then normalized per video:

$$G(c) = \frac{S(c) - S_{\min}}{S_{\max} - S_{\min}}, \quad (6)$$

where $S_{\max}$ and $S_{\min}$ are the highest and lowest scores among captions of the same video.

The normalized scores $G(c)$ serve as soft weights during training, enabling the model to benefit from diverse expressions while prioritizing more consistent and informative ones. The caption with the highest normalized score is selected to supervise the teacher network. Before supervision, this caption undergoes part-of-speech (POS) filtering via spaCy

[36] to extract noun tokens $T_n(c)$ and verb tokens $T_v(c)$, corresponding to object and action semantics, respectively. These token sets guide the teacher network in learning object-level features F_o and action-level features F_a .

3.4. Teacher network (T-Net)

The teacher network (T-Net) is designed to extract semantically rich and hierarchically structured video representations through four key components: (1) an object encoding block supervised by noun tokens, which captures spatial object-level features; (2) a motion encoding block utilizing multi-head self-attention to model temporal dynamics; (3) an action encoding block that fuses object and motion cues via cross-attention and is guided by verb tokens to capture action semantics; and (4) a global video encoding block that integrates hierarchical attention with a BiLSTM to model high-level contextual and temporal dependencies. Supervision for object and action features is guided by high-quality captions identified through the proposed caption grading mechanism. The decoder generates captions from the fused frame-level and global features, and the network is trained using a grade-weighted cross-entropy loss to emphasize informative annotations while mitigating the impact of noisy labels.

3.4.1. Object feature encoding

Frame-level features F_v are first embedded to obtain $E_v \in \mathbb{R}^{N \times d_e}$, which are linearly projected to produce object-level features:

$$F_o = E_v \cdot W_o + b_o, \quad (7)$$

where W_o and b_o are learnable parameters.

To supervise object semantics, F_o is fed through a fully connected layer and a pretrained BERT prediction head, yielding a distribution $p_o \in \mathbb{R}^{N \times |V|}$ over the vocabulary V . Noun tokens $T_n(c)$ are extracted from the highest-graded caption c , tokenized, and converted into a smoothed target distribution p_{noun} .

The training objective minimizes the KL divergence between p_o and p_{noun} :

$$L_{\text{KL,obj}} = \sum_{i=1}^{N_n} \sum_{v \in V} p_{\text{noun}}(v) \log \left(\frac{p_{\text{noun}}(v)}{p_o(v)} \right), \quad (8)$$

where $N_n = |T_n(c)|$. This guides the teacher to learn object features aligned with noun semantics from high-quality captions.

3.4.2. Motion feature encoding

The embedded frame features $E_v \in \mathbb{R}^{N \times d_e}$ are passed through six stacked multi-head self-attention (MHSA) layers to capture long-range temporal dependencies. The motion features F_m are computed as:

$$F_m = \text{FFN}(\text{MHSA}(Q, K, V)), \quad (9)$$

where $Q = W_{mq} E_v$, $K = W_{mk} E_v$, and $V = W_{mv} E_v$, with W_{mq} , W_{mk} , W_{mv} as learnable matrices. MHSA captures inter-frame relationships, and the FFN refines the resulting motion features.

3.4.3. Action feature encoding

Action features are derived using an Action Attention Block that integrates object features F_o and motion features F_m , as shown in Fig. 3(a). This block contains six stacked layers, each combining self-attention on object features and cross-attention with motion features to model fine-grained object-context interactions essential for action recognition.

Self-attention on object features: Object features F_o are refined using multi-head self-attention followed by a feed-forward network (FFN) to capture intra-object dependencies across frames. The output is:

$$F'_o = \text{FFN}(\text{MHSA}(F_o)), \quad (10)$$

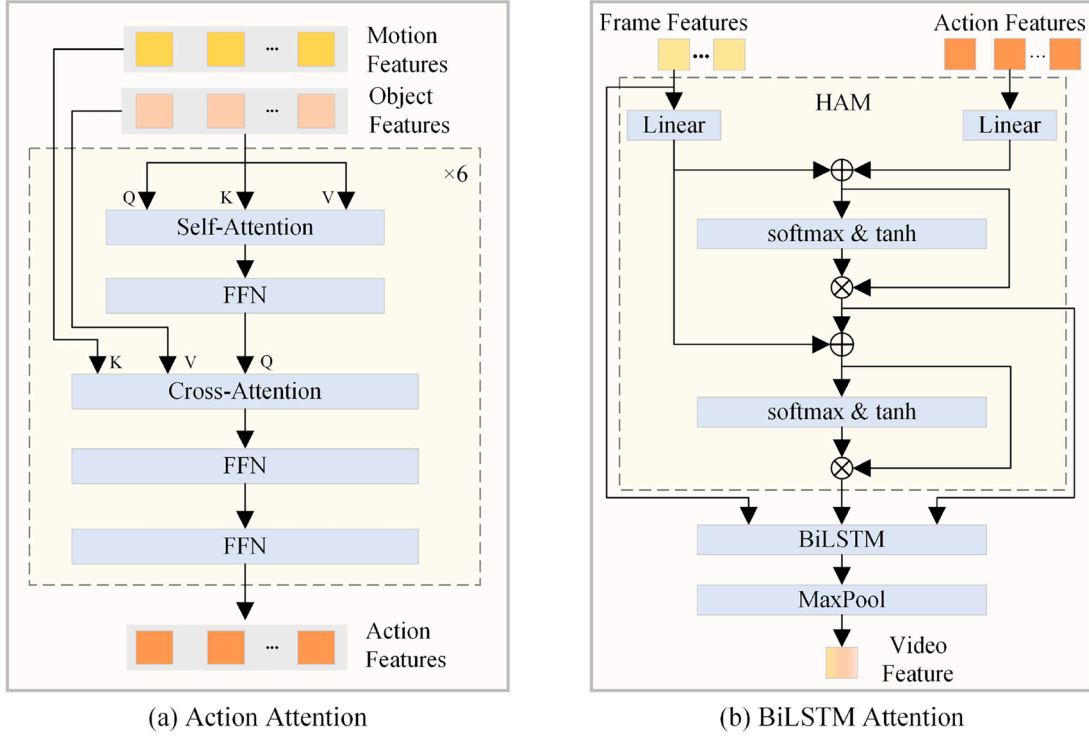


Fig. 3. Key components of the proposed teacher network for semantic feature extraction. (a) Action Attention Block: Fuses object features F_o and motion features F_m to generate action-level representations F_a . (b) BiLSTM Attention Layer: Captures long-range temporal dependencies across frames while integrating spatial and semantic cues.

where the FFN stabilizes and enriches the representations via normalization and non-linear transformations.

Cross-attention between object and motion features: To derive action features F_a , refined object features F'_o , original object features F_o , and motion features F_m are fused using multi-head cross-attention followed by stacked feed-forward networks:

$$F_a = \text{FFN}(\text{FFN}(\text{MHA}(Q, K, V))), \quad (11)$$

where $Q = W_{aq}F'_o$, $K = W_{ak}F_m$, and $V = W_{av}F_o$, with W_{aq}, W_{ak}, W_{av} as learnable projection matrices. This mechanism captures detailed spatiotemporal interactions crucial for action understanding.

Supervision of action features: Action features F_a are supervised using the verb token set $T_v(c)$, extracted from the highest-scoring caption c via caption grading. A predicted action distribution p_a is derived from F_a , while a smoothed target distribution p_{verb} is generated from $T_v(c)$, following the same approach as in object supervision.

Semantic alignment is enforced by minimizing the KL divergence between p_a and p_{verb} :

$$L_{\text{KL-act}} = \sum_{i=1}^{N_v} \sum_{v \in V} p_{\text{verb}}(v) \log \left(\frac{p_{\text{verb}}(v)}{p_a(v)} \right), \quad (12)$$

where $N_v = |T_v(c)|$. This loss encourages the action features to align with the verb semantics, enhancing contextual relevance and action understanding.

3.4.4. Video feature encoding

To capture global video semantics, the action features $F_a \in \mathbb{R}^{N \times d_e}$ and frame features $E_v \in \mathbb{R}^{N \times d_e}$ are fused using a Hierarchical Attention Mechanism (HAM) followed by a bidirectional LSTM (BiLSTM), as shown in Fig. 3(b). HAM models fine-grained action-frame interactions and contextual relationships between frames, while the BiLSTM captures long-term temporal dependencies. This combination yields a uni-

fied video representation that integrates spatial, temporal, and action-level cues.

Hierarchical attention mechanism: The hierarchical attention mechanism (HAM) is designed to capture both local action-frame interactions and global frame-level contextual dependencies. It consists of two sequential attention stages.

In the first stage, action features $F_a \in \mathbb{R}^{N \times d_e}$ and frame features $E_v \in \mathbb{R}^{N \times d_e}$ are projected into a shared embedding space using learnable matrices $U_a, U_v \in \mathbb{R}^{d_e \times d_e}$, with a bias term $b_a \in \mathbb{R}^{d_e}$. The interaction tensor is computed as:

$$A_1 \in \mathbb{R}^{N \times N \times d_e} = U_v E_v + U_a F_a + b_a, \quad (13)$$

where broadcasting aligns dimensions for addition. Specifically, $U_v E_v$ is expanded to $\mathbb{R}^{N \times 1 \times d_e}$ along the action dimension, and $U_a F_a$ is expanded to $\mathbb{R}^{1 \times N \times d_e}$ along the frame dimension. This forms pairwise action-frame interactions across all positions.

A non-linear activation function (tanh) followed by softmax normalization along the action dimension yields the attention weights:

$$W_1 \in \mathbb{R}^{N \times N} = \text{softmax}(\tanh(A_1)). \quad (14)$$

Here, $W_1[i, j]$ represents the normalized attention score indicating the contribution of the j -th action to the i -th frame. Hence, $W_1 \in \mathbb{R}^{N \times N}$ encodes a complete pairwise attention map between all frame and action positions.

These weights are used to aggregate action-frame interactions:

$$F_{\text{attn1}} \in \mathbb{R}^{N \times d_e} = W_1 \cdot A_1, \quad (15)$$

where $\cdot$ denotes element-wise multiplication followed by summation over the action dimension.

In the second stage, frame-to-frame contextual dependencies are modeled. The first attention output F_{attn1} is combined with the original frame features E_v , along with a new bias term $b_v \in \mathbb{R}^{d_e}$:

$$A_2 \in \mathbb{R}^{N \times N \times d_e} = U_v E_v + F_{\text{attn1}} + b_v. \quad (16)$$

Softmax is again applied to compute the frame-level attention weights:

$$W_2 \in \mathbb{R}^{N \times N} = \text{softmax}(\tanh(A_2)), \quad (17)$$

and these weights are used to refine contextual dependencies across frames:

$$F_{\text{attn2}} \in \mathbb{R}^{N \times d_e} = W_2 \cdot A_2. \quad (18)$$

Finally, the outputs from both stages, along with the original frame features, are concatenated to form the unified representation:

$$F_{\text{concat}} \in \mathbb{R}^{N \times 3d_e} = [E_v; F_{\text{attn1}}; F_{\text{attn2}}], \quad (19)$$

where $[\cdot; \cdot]$ denotes the concatenation operation along the feature dimension.

Temporal modeling with BiLSTM: The unified features $F_{\text{concat}} \in \mathbb{R}^{N \times 3d_e}$ are fed into a bidirectional LSTM to model global temporal dependencies:

$$O_{\text{BiLSTM}} \in \mathbb{R}^{N \times d_e} = \text{BiLSTM}(F_{\text{concat}}). \quad (20)$$

Max pooling across the temporal axis yields a compact global video representation:

$$F_{\text{vid}} \in \mathbb{R}^{1 \times d_e} = \text{maxpool}(O_{\text{BiLSTM}}). \quad (21)$$

This captures spatial, temporal, and action-aware semantics for downstream caption generation.

3.4.5. Caption decoding and generation

To generate captions $Y = \{y_1, y_2, \dots, y_T\}$, each word y_t is predicted based on the video context and previously generated words $Y_{<t}$. The frame features $E_v \in \mathbb{R}^{N \times d_e}$ and global video features $F_{\text{vid}} \in \mathbb{R}^{1 \times d_e}$ are concatenated to form $F_v \in \mathbb{R}^{(N+1) \times d_e}$, providing both local and global visual context.

The word history $Y_{<t}$ is embedded via an MLP to produce $F^T \in \mathbb{R}^{L \times d_e}$, where L is the sequence length. Combined with F_v and position embeddings F^X , the full input sequence is:

$$E = [F_v; F^T] + F^X, \quad F^X \in \mathbb{R}^{(N+1+L) \times d_e}. \quad (22)$$

Masked self-attention computes contextual representations:

$$h_t = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M_{\text{mask}}\right)V, \quad (23)$$

$$Q = W_{eq}E, \quad K = W_{ek}E, \quad V = W_{ev}E, \quad (24)$$

where W_{eq}, W_{ek}, W_{ev} are learnable matrices and M_{mask} prevents attention to future tokens. The next word y_t is predicted via a classification head:

$$y_t \sim P(y_t | Y_{<t}, F_v) = \text{Softmax}(h_t W_{\text{cls}}), \quad (25)$$

where $W_{\text{cls}} \in \mathbb{R}^{d_e \times |V|}$ projects to the vocabulary space.

This process iterates over t to produce a coherent caption describing the video.

3.5. Student network (S-Net) and hierarchical knowledge distillation

The student network (S-Net) adopts a lightweight architecture inspired by the teacher network (T-Net) but is optimized for computational efficiency. This optimization is achieved by simplifying the teacher network's structure: the self-attention block for motion feature encoding is removed, and only a single action attention layer is retained for action feature encoding. The objective of the student network is to achieve performance close to the teacher network while significantly reducing computational complexity. By leveraging hierarchical knowledge distillation, S-Net emulates the high-level semantic understanding and feature representations of T-Net.

During training, hierarchical supervision is applied at three levels: object feature alignment, action feature alignment, and word-level

alignment. This multi-level distillation framework ensures that the student network progressively approximates the semantic representations and output distributions of the teacher network.

Object feature alignment: The alignment of object features is performed between the object features F_o generated by the teacher network and F_{o_s} generated by the student network. These features are projected into semantic representations using fully connected layers and subsequently transformed into probability distributions p_o (teacher) and p_{o_s} (student) over the vocabulary V using a pretrained BERT prediction head. To align these distributions, the Kullback-Leibler (KL) divergence loss is applied:

$$L_{\text{distill_obj}} = \sum_{i=1}^N p_o^i \log \left(\frac{p_o^i}{p_{o_s}^i} \right), \quad (26)$$

This alignment ensures that the student network learns to encode object features in a manner consistent with the semantic representations of the teacher network.

Action feature alignment: To enable the student network to encode temporal and spatiotemporal dynamics effectively, a similar distillation process is applied to the action features F_a (teacher) and F_{a_s} (student). These features are projected into semantic representations and transformed into probability distributions p_a (teacher) and p_{a_s} (student) using fully connected layers and a pretrained BERT prediction head. The KL divergence loss for action features is defined as:

$$L_{\text{distill_act}} = \sum_{i=1}^N p_a^i \log \left(\frac{p_a^i}{p_{a_s}^i} \right). \quad (27)$$

By minimizing $L_{\text{distill_act}}$, the student network is guided to generate action representations that closely align with those of the teacher network, effectively capturing temporal and spatiotemporal semantics.

Word-level alignment: At the sequence level, hierarchical supervision extends to the word generation process. At each time step t , the word probability distribution $P(y_t)$ produced by the teacher network is aligned with the corresponding probability distribution $P_s(y_t)$ produced by the student network. The KL divergence loss is applied as follows:

$$L_{\text{distill_word}} = \sum_{y \in V} P(y_t) \log \left(\frac{P(y_t)}{P_s(y_t)} \right). \quad (28)$$

This alignment ensures that the word generation outputs of the student network are consistent with those of the teacher network, thereby reducing discrepancies in video captioning results.

By progressively aligning object, action, and word-level distributions, the student network learns to approximate the hierarchical understanding of the teacher network. Despite its simplified architecture, the student network achieves competitive performance with significantly lower computational complexity.

3.6. Loss function

To address noisy annotations in video captioning datasets, we propose a weighted cross-entropy (WCE) loss that incorporates ground-truth caption quality scores. This approach encourages the model to prioritize reliable captions during training. The WCE loss is defined as:

$$L_{\text{WCE}} = \sum_{j=1}^{n_i} G(c^j) \sum_{t=1}^T y_t^j \log P(y_t | Y_{<t}, F_v), \quad (29)$$

where n_i is the number of ground-truth captions for a video, T is the maximum sequence length, y_t^j is the true distribution of the t -th word in caption c^j , $G(c^j)$ is its normalized quality score, and $P(y_t | Y_{<t}, F_v)$ is the predicted probability of the word conditioned on previous words $Y_{<t}$ and video features F_v .

The total training loss integrates supervision and knowledge distillation components:

$$L = L_{WCE_t} + L_{WCE_s} + 0.5(L_{KL_{obj}} + L_{KL_{act}} + L_{distill_{obj}} + L_{distill_{act}} + L_{distill_{word}}), \quad (30)$$

where L_{WCE_t} and L_{WCE_s} are the WCE losses for the teacher and student networks, respectively. $L_{KL_{obj}}$ and $L_{KL_{act}}$ denote KL divergence losses that align object- and action-level features with semantic cues from ground-truth captions. $L_{distill_{obj}}$, $L_{distill_{act}}$, and $L_{distill_{word}}$ are hierarchical distillation losses that transfer object-level, action-level, and word-level knowledge from the teacher to the student network.

4. Experiments

4.1. Datasets and evaluation metrics

MSVD dataset: The MSVD dataset [11] contains 1970 short YouTube video clips, each annotated with approximately 40 English captions. Following standard practice, we use 1200 videos for training, 100 for validation, and 670 for testing.

MSR-VTT dataset: The MSR-VTT dataset [12] consists of 10,000 open-domain YouTube videos, each paired with 20 English captions. Compared to MSVD, it presents greater challenges due to its larger scale, broader category diversity, and more complex visual scenes. We follow the standard split: 6,573 videos for training, 497 for validation, and 2990 for testing. **VATEX dataset:** The VATEX dataset [13] is a large-

scale multilingual benchmark with over 41,250 videos, each annotated with 10 English and 10 Chinese captions. The videos cover a broad range of everyday human activities and diverse scenes. We adopt the official

split: 25,991 for training, 3000 for validation, and 6000 for testing (public test set).

Evaluation metrics: We evaluate caption quality using BLEU-4 [28], METEOR [37], ROUGE-L [38], and CIDEr [27]. BLEU-4 and CIDEr focus on n-gram precision and fluency, while METEOR and ROUGE-L assess semantic alignment and structural quality. Higher scores indicate better caption accuracy and relevance.

4.2. Implementation details

Preprocessing of visual features and captions: An average of 15 frames per video was extracted using the CLIP multimodal pre-trained model, projecting features into a 768-dimensional space. Visual features and ground-truth captions were preprocessed offline without participating in network training.

Ground-truth captions were truncated to a maximum length of 22 words, converted to lowercase, and stripped of punctuation. The vocabulary set V used in our experiments is defined by the tokenizer shared between CLIP and BERT, resulting in a vocabulary size of $|V| = 49,408$.

Training details: The model was trained using the Adam optimizer. The initial learning rate was set to 7×10^{-5} for the MSR-VTT dataset and 1×10^{-4} for the MSVD dataset, with a batch size of 64 over 20 epochs.

4.3. Performance and efficiency comparison with state-of-the-art methods

We evaluate CapDistill on three widely used benchmarks: MSVD, MSR-VTT, and VATEX, and compare it with recent state-of-the-art video captioning methods. The results are summarized in Tables 1 and 2.

On MSVD, CapDistill achieves a CIDEr score of 127.0 and ranks second on three metrics and third on one metric. It improves CIDEr by 16.7 points over CoCap, a strong CLIP-based baseline. While its performance is slightly lower than Track4Cap, CapDistill requires fewer

Table 1

Performance comparison of CapDistill and state-of-the-art methods on MSVD and MSR-VTT. G, IRV2, IV4, R101, R152, 3D-RX, VidSwin, and FR denote GoogLeNet, Inception-ResNet-V2, InceptionV4, ResNet-101, ResNet-152, 3D ResNeXt, Video Swin Transformer, and Faster R-CNN, respectively. Best, second-best, and third-best results are marked in underline, *italic*, and **bold**. Metrics include BLEU-4 (B4), METEOR (M), ROUGE-L (R), CIDEr (C), parameters (P), FLOPs (F), and inference time (T), measured on an NVIDIA 2080 Ti (12GB).

Method	Feature	Complexity			MSVD				MSR-VTT			
		P(M)↓	F(G)↓	T(s)↓	B4	M	R	C	B4	M	R	C
Res-F2F ₁₉ [14]	G + R101 + R152	–	–	–	52.4	35.7	–	84.3	41.4	29.0	61.3	48.9
SAAT ₂₀ [19]	IRV2 + C3D + FR	29	4	0.03	46.5	33.5	69.4	81.0	39.9	27.7	61.2	51.0
STG-KD ₂₀ [31]	R101 + I3D + FR	–	60	–	52.2	36.9	73.9	93.0	40.5	28.3	60.9	47.1
ORG-TRL ₂₀ [32]	IRV2 + C3D + FR	–	–	0.25	54.3	36.4	73.9	95.2	43.6	28.8	62.1	50.9
DRPN ₂₁ [10]	ECO + 3D-RX	–	–	–	49.2	34.2	71.5	86.4	39.5	27.7	61.0	49.2
SGN ₂₁ [39]	R101 + C3D	15	1.05	0.094	52.8	35.5	72.9	94.3	40.8	28.3	60.8	49.5
OpenBook ₂₁ [2]	IRV2 + 3D-RX	–	–	–	–	–	–	–	43.6	28.8	62.1	50.9
TTA ₂₁ [20]	IRV2 + C3D + FR	–	–	–	51.8	35.5	72.4	87.7	41.4	27.7	61.1	46.7
HRNAT ₂₂ [40]	IRV2 + I3D	53	2.73	0.056	55.7	36.8	74.1	98.1	42.1	28.0	61.6	48.2
TVRD ₂₂ [41]	IRV2 + C3D + FR	226	190.3	–	50.5	34.5	71.7	84.3	43.0	28.7	62.2	51.8
SEM-POS ₂₃ [42]	IRV2 + C3D	–	–	–	60.1	38.5	76.0	108.3	45.2	30.7	<u>64.1</u>	53.1
RSFD ₂₃ [21]	R101 + 3D-RX	29	1.65	102.8	51.2	35.7	72.9	96.7	43.4	29.3	62.3	53.1
MesNet ₂₃ [4]	IV4 + C3D	48	–	<u>0.002</u>	53.7	36.6	74.3	102.1	42.0	28.6	61.0	50.1
HTG + HMG ₂₃ [8]	R152 + C3D + FR	–	–	–	52.7	35.2	72.8	91.4	42.1	28.4	61.6	48.9
MAN ₂₄ [16]	R101 + 3D-RX	857	35.3	–	59.7	37.3	74.3	101.5	41.3	28.0	61.4	49.8
ALSO-Net ₂₄ [25]	R101 + 3D-RX	–	–	–	55.7	35.9	73.0	89.0	41.9	28.9	62.0	51.3
SNCL ₂₄ [22]	IRV2 + C3D + FR	–	–	–	59.7	38.9	77.2	106.3	43.8	29.6	63.6	54.6
GSEN ₂₄ [17]	IRV2 + I3D + FR	<u>4.8</u>	–	0.120	58.8	37.6	75.2	102.5	42.9	28.4	61.7	51.0
KG-VCN ₂₅ [9]	IRV2 + C3D + FR	–	–	–	<u>64.9</u>	39.7	77.2	107.1	45	28.7	62.5	51.9
SwinBERT ₂₂ [6]	VidSwin	225	1154	0.498	55.7	39.6	75.7	109.4	41.9	29.8	62.1	53.7
VASTA ₂₂ [5]	VidSwin	–	–	–	56.1	39.1	76.5	106.4	44.2	30.2	62.9	56.1
EMCL-Net ₂₂ [30]	CLIP(ViT/B-32)	–	–	–	–	–	–	–	<u>45.3</u>	30.2	63.2	54.6
CoCap ₂₃ [26]	CLIP(ViT/L-14)	441	103	0.259	59.1	39.5	76.5	110.3	<u>44.4</u>	30.3	63.4	57.2
IVRC ₂₄ [15]	CLIP(ViT/B-32)	–	–	–	58.8	40.3	77.4	116.0	43.7	30.2	63.0	57.1
Track4Cap ₂₅ [18]	CLIP(ViT/L-14)	91	1.3	0.022	<u>62.1</u>	<u>42.5</u>	<u>79.8</u>	<u>127.2</u>	44.6	30.5	63.6	<u>57.7</u>
Ours(CapDistill)	CLIP(ViT/L-14)	84	<u>0.7</u>	<i>0.010</i>	61.6	<i>42.4</i>	<i>79.1</i>	<i>127</i>	<u>45.3</u>	30.2	63.8	<u>57.9</u>

Table 2

Performance comparison of CapDistill with state-of-the-art (SOTA) methods on the VATEX dataset. The best, second-best, and third-best results are highlighted in underline, *italic*, and **bold**, respectively.

Method	Feature	Complexity			VATEX			
		P(M)↓	F(G) ↓	T(s) ↓	B4	M	R	C
VATEX ₁₉ [13]	I3D	<u>35</u>	–	–	28.4	21.7	47.0	45.1
ORG-TRL ₂₀ [32]	IRV2 + C3D + FR	–	–	0.25	32.1	22.2	48.9	49.7
HRNAT ₂₂ [40]	IRV2 + I3D	53	2.73	<i>0.056</i>	32.5	22.3	49.0	50.7
SwinBERT ₂₂ [6]	VidSwin	225	1154	0.498	<u>37.8</u>	<u>26.1</u>	<u>53.0</u>	<u>71.6</u>
CARE ₂₃ [24]	CLIP(ViT/B-32) + 3D-RX	–	–	–	–	20.1	43.6	45.0
CoCap ₂₃ [26]	CLIP(ViT/L-14)	441	103	0.259	31.4	23.2	49.4	54.4
IVRC-XE ₂₄ [15]	CLIP(ViT/B-32)	–	–	–	31.2	23.8	49.4	54.4
IVRC-DXE ₂₄ [15]	CLIP(ViT/B-32)	–	–	–	32.8	<i>24.0</i>	<i>50.3</i>	<i>57.4</i>
Ours(CapDistill)	CLIP(ViT/B-32)	84	<u>0.7</u>	<u>0.010</u>	31.6	23.9	49.6	55.3

parameters (84M vs. 91M), lower FLOPs (0.7G vs. 1.3G), and faster inference (0.010s vs. 0.022s). This trade-off is relevant for applications with strict resource constraints. The difference in performance may be attributed to MSVD's relatively simple scenes and short captions, where Track4Cap's object tracking and relation modeling are advantageous.

On MSR-VTT, CapDistill achieves the highest CIDEr (57.9) and BLEU-4 (45.3) scores. It ranks first on two metrics and second on one metric, outperforming both CoCap and Track4Cap, with CIDEr gains of 0.7 and 0.2 and BLEU-4 improvements of 0.9 and 0.7, respectively. CapDistill maintains the lowest FLOPs and the second fastest inference time among all compared models, including Track4Cap (1.3G FLOPs, 0.022s) and SGN (1.05G FLOPs, 0.094s).

On the VATEX dataset, CapDistill ranks second among CLIP-based methods, achieving a CIDEr score of 55.3, ROUGE-L of 49.6, and ME-TOR of 23.9. It also maintains the lowest computational cost (0.7G FLOPs) and the fastest inference time (0.010s) among top-performing models. In contrast, SwinBERT attains the highest accuracy but requires substantially more computation (1154G FLOPs and 0.498s per inference) due to its use of full 3D video modeling and large-scale transformer decoding.

In summary, CapDistill provides a favorable balance between accuracy and efficiency. Compared to other recent methods, it achieves the lowest FLOPs, the second fastest inference time, and relatively low parameter counts while delivering competitive or superior accuracy across all evaluated benchmarks. By relying on lightweight frame-level representations and hierarchical distillation, CapDistill enables real-time captioning in resource-constrained environments while preserving high semantic fidelity and generalization performance.

4.4. Qualitative analysis

As shown in Fig. 4, we present qualitative examples of video captions generated by the proposed CapDistill method, compared with three state-of-the-art models: SAAT [19], STG-KD [31], and Track4Cap [18]. SAAT and STG-KD incorporate distillation-based learning, while Track4Cap represents a more recent and efficient relation-aware captioning model. The evaluation is conducted on the MSR-VTT dataset to assess each model's ability to generate accurate and contextually rich descriptions.

The results indicate that CapDistill produces more precise and context-aware captions than the other compared methods. While SAAT effectively identifies primary actions, its outputs are often brief and lack descriptive detail. STG-KD and Track4Cap generate more fluent captions but still struggle to capture complex scene context and multi-object interactions.

In contrast, CapDistill captures both object and action semantics in a more comprehensive manner. For instance, in the bottom-right video of Fig. 4, SAAT, STG-KD, and Track4Cap describe the scene as a man and woman kissing but omit the baby's reaction—a key contextual element. CapDistill successfully includes this detail, demonstrating its ability to model deeper narrative context and subtle visual relationships.

These results highlight the strength of CapDistill's hierarchical knowledge distillation strategy, which transfers multi-level semantics—including object- and action-level cues—from a powerful teacher model into a lightweight student network. This enables the generation of contextually richer and semantically complete captions, even in complex or nuanced scenarios.



GroundTruth: female models are walking down a runway in dresses.

SAAT: a women is walking.

STG-KD: a woman is walking on a stage.

Track4Cap: two women is walking on a stage.

CapDistill: a women in dress are walking on a stage.



GroundTruth: two women are standing with dress and cooking a food item.

SAAT: two women are cooking dishes.

STG-KD: two women in dress are cooking food.

Track4Cap: two women are showing how to cook food.

CapDistill: two women in blue shirt are cooking dishes in kitchen.



GroundTruth: a soccer team is celebrating for a victory.

SAAT: women are celebrating.

STG-KD: women are kissing and hugging.

Track4Cap: women are celebrating and hugging.

CapDistill: women are celebrating a goal.



GroundTruth: a man is kissing a woman and the baby is smiling.

SAAT: a man and a woman are kissing.

STG-KD: a man is kissing a woman in a room.

Track4Cap: a man and a woman are kissing in a room.

CapDistill: a man and a woman are kissing while baby smiling.

Fig. 4. Examples of video captions generated on the MSR-VTT dataset by SAAT, STG-KD, Track4Cap, and the proposed CapDistill method.

Table 3

Ablation study on the MSVD and MSR-VTT datasets evaluating the effectiveness of the hierarchical distillation strategies, including teacher-student hierarchical distillation (TSHD) and caption-to-teacher hierarchical distillation (CTHD).

Method	MSVD				MSR-VTT			
	B4	M	R	C	B4	M	R	C
(a) S-Net for inference								
S-Net-only	58.2	38.2	76.5	119.7	43.1	27.8	61.1	55.4
TSHD	60.8	41.2	78.2	122.4	43.8	28.5	62.4	56.1
TSHD + CTHD (CapDistill)	61.6	42.4	79.1	127.0	45.3	30.2	63.8	57.9
(b) T-Net for inference								
T-Net-Only	61.5	41.5	78.8	124.2	44.8	29.4	63.4	57.2
T-Net + CTHD	62.5	42.6	79.8	128.1	45.6	30.5	64	58.6

Table 4

Feature ablation on MSVD and MSR-VTT assessing the contribution of frame, object, action, and video-level features in CapDistill.

Feature				MSVD				MSR-VTT			
Frame	Object	Action	Video	B4	M	R	C	B4	M	R	C
✓	×	✓	✓	62.4	41.8	79.4	125.7	43.0	30.0	62.8	55.5
✓	✓	×	✓	62.4	42.4	79.5	125.5	43.0	30.0	62.7	55.9
✓	✓	✓	✓	61.2	41.9	79.0	124.1	42.6	30.1	62.7	56.1
✓	×	×	✓	61.6	42.4	79.1	127.0	45.3	30.2	63.8	57.9

4.5. Ablation study

4.5.1. Ablation study on distillation strategies and feature combinations

To assess the effectiveness of our hierarchical distillation design, we perform ablation studies on two aspects: (1) the contribution of the two-stage distillation mechanisms, including teacher-student hierarchical distillation (TSHD) and caption-to-teacher hierarchical distillation (CTHD); and (2) the impact of different feature combinations on captioning performance.

We compare five model variants. The S-Net-only model uses the student network trained without any distillation. The TSHD variant applies teacher-student distillation without caption-guided supervision. The full model CapDistill incorporates both TSHD and CTHD, using only the student network at inference. For reference, we also evaluate the teacher network in two settings: T-Net-only (without caption supervision) and T-Net+CTHD (with caption-guided hierarchical distillation only).

As shown in Table 3, the S-Net-only variant performs the worst, indicating that the student network alone lacks sufficient capacity. Introducing TSHD leads to significant improvements by transferring rich semantics and temporal context from the teacher. Adding CTHD further enhances results, with over 1-point gains across most metrics, demonstrating the benefit of fine-grained supervision from ground-truth captions.

The T-Net+CTHD model outperforms the T-Net-only baseline, highlighting that caption-guided supervision enhances the teacher's representations. Importantly, the student in the full CapDistill setup matches

or exceeds the teacher's performance, validating the success of dual hierarchical distillation in transferring knowledge effectively.

We further analyze the effect of combining different feature types, including frame, object, action, and video-level features. As shown in Table 4, the combination of frame and video-level features yields the best performance. In contrast, incorporating all four types leads to degradation, likely due to redundant or overlapping information. Frame features capture fine-grained spatial details, while video-level features provide global spatiotemporal context. Adding object and action features appears to reduce generalization and increase the risk of overfitting.

4.5.2. Effect of noun and verb supervision in caption-to-teacher distillation

We conduct ablation experiments on the MSVD and MSR-VTT datasets to evaluate the effectiveness of hierarchical caption-to-teacher distillation under different supervision strategies. Results are presented in Table 5.

Without caption-guided supervision, both the teacher and student networks achieve the lowest performance. Applying noun-level supervision (L_{KL_obj}) improves object feature learning and leads to clear gains. Verb-level supervision (L_{KL_act}) also boosts performance by enhancing action understanding, though the improvement is generally smaller than that from noun-level supervision, except for the student network on MSR-VTT.

The best results are obtained when both noun- and verb-level losses are combined. This demonstrates the complementary effect of object and action semantics, enabling the teacher to learn richer representa-

Table 5

Comparison of caption-to-teacher distillation strategies, including noun-only, verb-only, and combined supervision. Results are reported for both teacher and student networks on the MSVD and MSR-VTT datasets.

Strategy		T-Net								S-Net							
		MSVD				MSR-VTT				MSVD				MSR-VTT			
KL _{obj}	KL _{act}	B4	M	R	C	B4	M	R	C	B4	M	R	C	B4	M	R	C
×	×	61.5	41.5	78.8	124.2	44.8	29.4	63.4	57.2	60.8	41.2	78.2	122.4	43.8	28.5	62.4	56.1
✓	×	62.1	42	79.5	126.4	45	29.9	63.6	58	60.8	41.5	78.8	124.1	44.2	28.5	62.8	56.6
×	✓	61.8	41.7	79	125.4	45.2	29.7	63.7	57.7	61.2	41.3	78.3	123.5	44.4	29.1	63.2	56.4
✓	✓	62.5	42.6	79.8	128.1	45.6	30.5	64	58.6	61.6	42.4	79.1	127	45.3	30.2	63.8	57.9

Table 6

Impact of different weighting schemes for noun- and verb-level supervision on the performance of the teacher network (T-Net) across MSVD and MSR-VTT datasets.

Weighting Scheme		T-Net (w/o S-Net)							
		MSVD				MSR-VTT			
KL_obj	KL_act	B4	M	R	C	B4	M	R	C
0	0	61.5	41.5	78.8	124.2	44.8	29.4	63.4	57.2
0.1	0.9	61.4	41.8	79.5	128.4	43.8	30.3	63.1	57.1
0.3	0.7	62.1	42.1	80.0	129.3	44.7	30.3	63.4	57.6
0.7	0.3	61.6	41.8	79.5	129.0	44.9	30.2	63.3	57.9
0.9	0.1	61.3	42.0	79.5	127.8	43.8	30.5	63.5	57.1
0.5	0.5	62.5	42.6	79.8	128.1	45.6	30.5	64	58.6

tions and transfer knowledge more effectively. As a result, the student produces more accurate and context-aware captions.

We further evaluate the effect of different weighting strategies for noun and verb supervision. As shown in Table 6, equal weighting (0.5 for each loss) yields the best overall performance, suggesting a balanced contribution from both object- and action-level semantics is essential for optimal teacher representation learning.

4.5.3. Ablation study on hierarchical distillation from teacher to student

We conducted ablation experiments to evaluate the effectiveness of hierarchical teacher-student distillation on student network performance. Specifically, we examine four strategies: object-level semantics (Distill_obj), action-level semantics (Distill_act), word prediction (Distill_word), and global video feature alignment (Distill_global), which aligns teacher and student video features F_{vid} . The results are shown in Table 7.

Each strategy improves the student's performance, especially in CIDEr. Among them, word-level distillation (Distill_word) provides the largest gain, underscoring the importance of language-guided su-

pervision. Object-level supervision ranks second, while global alignment contributes the least, likely due to its indirect connection with caption semantics.

Combining object-level, action-level, and word-level distillation yields the best performance on both datasets, outperforming configurations that include Distill_global. This suggests that supervision through global video features may introduce redundancy that hinders effective learning.

To visualize these effects, Fig. 5 compares the captions generated using different strategies on the MSVD dataset. Object and word-level distillation leads to more accurate nouns, while action and word-level distillation improves verb precision. The full combination results in the most contextually rich and semantically aligned captions.

We further investigate the sensitivity of the model to the relative weights of Distill_obj, Distill_act, and Distill_word. As shown in Table 8, the equal weighting (0.5: 0.5: 0.5) achieves the best overall results, indicating that a balanced contribution from each distillation path is optimal for effective student learning.

4.5.4. Ablation of weighted cross-entropy loss and caption scoring metrics

We conduct ablation studies to assess the impact of the proposed caption evaluation method and the weighted cross entropy (WCE) loss on training effectiveness. Results are summarized in Table 9. In these experiments, CE denotes the standard cross entropy loss, and WCE refers to the proposed weighted variant. The Metric column indicates the evaluation metric used to score ground truth captions during training.

Across all datasets and metrics, WCE consistently outperforms CE by emphasizing high-quality captions and reducing the impact of noisy annotations.

Among the evaluated metrics, CIDEr provides the most effective supervision, outperforming both individual metrics and the combined average of BLEU-4, METEOR, ROUGE-L, and CIDEr. Its focus on n-gram consensus and redundancy penalization offers stronger training signals, leading to more accurate and diverse caption generation.

Table 7

Performance evaluation of different hierarchical distillation strategies on the MSVD and MSR-VTT datasets, including object-level, action-level, global feature, and word-level distillation.

Strategy				MSVD				MSR-VTT			
Distill_obj	Distill_act	Distill_global	Distill_word	B4	M	R	C	B4	M	R	C
×	×	×	×	58.2	38.2	76.5	119.7	43.1	27.8	61.1	55.4
✓	×	×	×	58.6	38.9	77.6	123.5	43.5	28.5	62.1	56.2
×	✓	×	×	58.4	38.7	77.2	123.8	43.3	28.2	61.8	56.3
×	×	✓	×	58.8	39	77.3	122.2	43.1	28.2	61.5	56.1
×	×	×	✓	60.4	41	77.9	124.5	44.5	29.3	62.7	56.3
✓	×	×	✓	60.8	41.2	78.5	125.1	45.0	29.6	63	56.5
×	✓	×	✓	59.5	40.8	77.8	124.6	44.5	29.1	62.4	56.2
×	×	✓	✓	60.1	40.4	77.1	123.5	44.4	29.2	62.1	55.7
✓	✓	×	✓	61.6	42.4	79.1	127.0	45.3	30.2	63.8	57.9
✓	×	✓	✓	61.5	41.4	78.5	125.2	44.5	29.5	63.2	55.5
×	✓	✓	✓	61.4	41.5	78.2	125.4	43.8	29.4	62.7	55.6
✓	✓	✓	✓	61.4	41.6	78.4	126.1	44.5	29.7	63.1	56.4



GroundTruth: a man is playing guitar in a band.

Distill_obj: a **man** is playing the **guitar**.

Disstill_act: a man is **playing** the violin.

CapDistill: the band is playing while the man is playing the guitar.



GroundTruth: a monkey wearing headphones is sitting on the bus.

Distill_obj: the **monkey** is listening with **headphones**.

Disstill_act: a monkey is **sitting** on a bus.

CapDistill: a monkey with headphones is sitting on a bus.

Fig. 5. Visualization results on the MSVD dataset. Distill_obj indicates the use of object semantic information distillation along with word prediction distillation, Distill_act refers to the use of action semantic information distillation along with word prediction distillation, and CapDistill represents the combined use of object, action, and word prediction distillation.

Table 8

Impact of different weighting schemes for object-, action-, and word-level distillation losses (Distill_obj, Distill_act, and Distill_word) on S-Net performance.

Weighting Scheme			S-Net (Base on T-Net's Best)							
			MSVD				MSR-VTT			
Distill_obj	Distill_act	Distill_word	B4	M	R	C	B4	M	R	C
0	0	0	58.2	38.2	76.5	119.7	43.1	27.8	61.1	55.4
0.9	0.3	0.3	61.1	41.9	78.9	126.1	44.6	30.0	64.4	57.4
0.3	0.9	0.3	60.1	41.7	79.2	125.1	44.4	30.1	63.5	57.3
0.3	0.3	0.9	62.9	42.2	79.6	126.7	42.7	30.2	62.8	57.3
0.7	0.7	0.1	61.1	41.8	78.9	125.8	44.5	29.7	63.1	56.4
0.5	0.5	0.5	61.6	42.4	79.1	127.0	45.3	30.2	63.8	57.9

Table 9

Performance comparison of standard Cross-Entropy (CE) and Weighted Cross-Entropy (WCE) losses combined with different caption evaluation metrics. *Average* refers to the mean of BLEU-4 (B4), METEOR (M), ROUGE-L (R), and CIDEr (C) scores.

Training	Metric	MSVD				MSR-VTT			
		B4	M	R	C	B4	M	R	C
CE	–	57.6	37.5	75.5	118.4	42.2	27.1	60.2	54.4
WCE	B4	61.7	40.8	78.4	125.4	44.4	29.3	63.2	57.6
	M	60.4	40.6	78.5	123.7	45	29.6	62.7	57.1
	R	59.8	39.8	78.7	122.4	45.1	29.4	63.4	57.4
	C	61.6	42.4	79.1	127.0	45.3	30.2	63.8	57.9
	Average	62.7	42.4	80.0	126.9	43.8	30.4	62.7	57.1

Table 10

Performance comparison using different numbers of attention layers in the teacher network. For each configuration with n layers, both the self-attention and action attention blocks contain n stacked layers.

No. of Attention Layers	MSVD				MSR-VTT			
	B4	M	R	C	B4	M	R	C
2	60.5	41.5	78.1	124.7	42.4	28.8	62.7	56.3
4	60.8	41.8	78.9	124.3	43.3	30.3	63.0	56.4
6	61.6	42.4	79.1	127.0	45.3	30.2	63.8	57.9
8	61.3	42.4	78.8	125.9	44.2	29.6	63.4	57.2
10	60.7	41.5	79	124.7	43.4	30.1	62.8	56.8
12	61.2	41.3	78.3	124.4	42.6	30.2	62.9	56.4

4.5.5. Ablation of the number of attention layers in the teacher network

This ablation study evaluates how varying the number of stacked attention layers in the teacher network's self-attention and action attention blocks affects the student network's performance. The goal is to determine the optimal teacher network depth for effective hierarchical knowledge distillation.

Increasing the number of attention layers enhances feature representation quality, enabling better knowledge transfer. However, as shown in Table 10, excessive layers introduce complexity, leading to overfitting and reduced distillation effectiveness.

Experimental results on the MSVD and MSR-VTT datasets demonstrate that using six attention layers in the teacher network yields the best student network performance. This configuration balances feature richness and computational efficiency. Fewer than six layers result in

underfitting due to limited feature extraction, while more than six layers cause overfitting, diminishing performance gains.

These findings highlight the importance of configuring the teacher network's depth to maximize the effectiveness of hierarchical knowledge distillation while ensuring computational efficiency.

4.6. Representative failure cases and limitations

Despite the effectiveness of the proposed model in generating video captions, several limitations remain that warrant further improvement. Representative examples of failed cases are shown in Fig. 6. Because we use a pre-trained CLIP encoder to extract frame-level features instead of a more complex 3D video encoder, the model sometimes struggles to recognize fine-grained action seman-

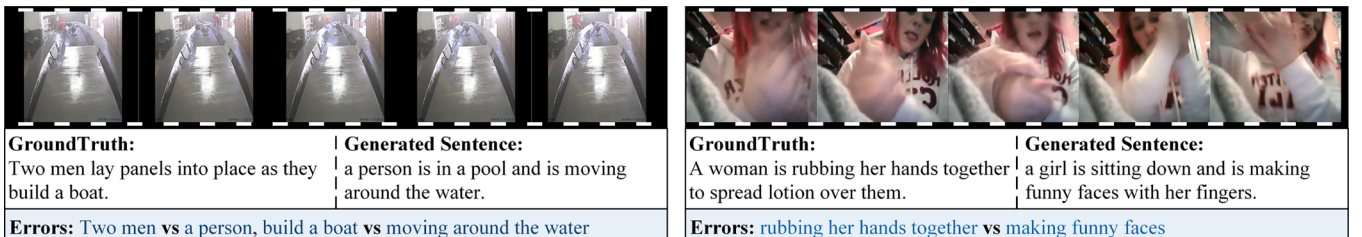


Fig. 6. Representative examples of failure cases on the VATEX dataset.

tics when key objects are small, partially occluded, or visually ambiguous.

For example, in the left case of Fig. 6, two men are building a boat, but they occupy only a small region of the frame, while the large, glare-covered surface of the boat resembles water. As a result, the model fails to correctly identify the building activity. In the right case, a woman is rubbing her hands to spread lotion, but the lotion bottle is largely occluded by her hand, which leads to inaccurate recognition of the action.

These examples illustrate that the model's reliance on frame-level features makes it sensitive to object scale, occlusion, and visual ambiguity. Future work could explore integrating temporal modeling or region-focused encoders to better capture small or partially hidden objects involved in actions.

5. Conclusion and future work

This paper presented CapDistill, a dual-hierarchical knowledge distillation framework designed to improve the efficiency and semantic richness of video captioning models. By combining caption-to-teacher and teacher-to-student distillation, the proposed approach enables a lightweight student network to learn multi-level semantic representations under strong supervision while maintaining low computational cost. A key strength of CapDistill is its robustness to noisy annotations, facilitated by a caption grading mechanism that emphasizes high-quality captions during training.

Extensive experiments on the MSVD, MSR-VTT, and VATEX datasets demonstrated that CapDistill achieves competitive or superior performance compared to recent state-of-the-art methods while significantly reducing inference time and FLOPs. Ablation studies further confirmed the effectiveness of each component, including the dual-hierarchical distillation stages and the caption evaluation mechanism, in improving caption quality and training stability.

Nevertheless, several limitations remain. As illustrated by representative failure cases, the reliance on CLIP-derived frame-level features can lead to inaccuracies when key objects are small, heavily occluded, or visually ambiguous.

Overall, the framework offers a practical solution for researchers and practitioners seeking efficient video captioning in real-world applications. Future work will explore integrating large language models as decoders to further enhance fluency and reasoning capabilities, extending the distillation process to incorporate richer linguistic signals such as adjectives and relational phrases, and leveraging additional modalities including subtitles and audio. Moreover, we plan to investigate region-focused visual encoders and temporal modeling techniques to better capture fine-grained actions and mitigate the observed failure cases, as well as develop domain adaptation strategies to improve generalization in low-resource or visually diverse scenarios.

CRedit authorship contribution statement

HuiLan Luo: Writing – review & editing, Methodology; **SiQi Wan:** Writing – original draft, Visualization, Software, Methodology; **Xia Cai:** Supervision, Software, Methodology, Data curation; **ChanJuan Wang:** Resources, Investigation, Formal analysis; **HongKun Chen:** Resources, Investigation, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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